US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12415242
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Kooiker; Wes et al.

System and apparatus for waste material collection from cutting machine

Abstract

A system for collecting and disposing waste from a material cutting machine may include a tray apparatus at least partially insertable into the cutting machine. The tray apparatus has an interior and an open top for receiving waste from the cutting machine. The tray apparatus may have a collection condition for holding waste in the interior and a disposal condition for dumping waste from the interior. The tray apparatus may include a base structure and a pan structure movably mounted on the base structure to move between a collection position and a disposal position. The tray apparatus may define a pair of fork pockets with each fork pocket being configured to receive a portion of a lift fork to facilitate lifting of the tray apparatus by equipment having lift forks.

Inventors: **Kooiker; Wes (Doon, IA), Vande Kamp; Justin L. (Hudson, SD), Koenen; Chad (Rock Valley, IA)**

Applicant: **Kooiker; Wes** (Doon, IA); **Vande Kamp; Justin L.** (Hudson, SD); **Koenen; Chad** (Rock Valley, IA)

Family ID: **1000006407493**

Assignee: **Kooima Company, Inc (Rock Valley, ME)**

Appl. No.: **17/749396**

Filed: **May 20, 2022**

Publication Classification

Int. Cl.: **B23Q11/00** (20060101)

U.S. Cl.:

CPC **B23Q11/0067** (20130101);

Field of Classification Search

CPC: B23Q (11/0042); B23Q (11/0067); B65D (88/56); B65G (65/23); B66F (9/19); B65F (1/122); B65F (3/26)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3430767	12/1968	Skirvin	210/167.01	B23Q 11/0042
9902558	12/2017	Genter	N/A	B65D 88/56

Primary Examiner: Kirsch; Andrew T

Attorney, Agent or Firm: Woods, Fuller, Shultz & Smith, PC

Background/Summary

BACKGROUND
Field
(1) The present disclosure relates to waste material collection apparatus and more particularly pertains to a new system and apparatus for waste material collection from a cutting machine which may also facilitate disposal of material from the apparatus.
SUMMARY
(2) In one aspect, the present disclosure relates to a system for collecting and disposing waste from a material cutting machine. The system may comprise a waste collection and disposal tray apparatus at least partially insertable into the material cutting machine to receive descending waste. The tray apparatus may have an interior for receiving waste from the material cutting machine, and an open top and a bottom opposite of the open top. The tray apparatus may have a collection condition characterized by the tray apparatus being configured to hold waste and a disposal condition characterized by the tray apparatus being configured to dump waste from the tray apparatus. The tray apparatus may comprise a base structure and a pan structure. In embodiments, the pan structure may be movably mounted on the base structure to move between a collection position corresponding to the collection condition and a disposal position corresponding to the disposal condition. In embodiments, the tray apparatus may define a pair of fork pockets with each fork pocket being configured to receive a portion of a lift fork to facilitate lifting of the tray apparatus by equipment having lift forks.
(3) There has thus been outlined, rather broadly, some of the more important elements of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional elements of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.
(4) In this respect, before explaining at least one embodiment or implementation in greater detail, it is to be understood that the scope of the disclosure is not limited in its application to the details of construction and to the arrangements of the components set forth in the following description or illustrated in the drawings. The disclosure is capable of other embodiments and implementations and is thus capable of being practiced and carried out in various ways. Also, it is to be understood

that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting.
(5) As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods and systems for carrying out the several purposes of the present disclosure. It is important, therefore, that the claims be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the present disclosure.
(6) The advantages of the various embodiments of the present disclosure, along with the various features of novelty that characterize the disclosure, are disclosed in the following descriptive matter and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure will be better understood and when consideration is given to the drawings and the detailed description which follows. Such description makes reference to the annexed drawings wherein:
(2) FIG. **1** is a schematic perspective view of a new system and apparatus for waste material collection from a cutting machine according to the present disclosure.
(3) FIG. **2** is a schematic upper perspective view of a tray apparatus of the system in the collection condition with the pan structure in the collection position, according to an illustrative embodiment.
(4) FIG. **3** is a schematic lower perspective view of the tray apparatus in the collection condition with the pan structure in the collection position, according to an illustrative embodiment.
(5) FIG. **4** is a schematic first perspective view of the tray apparatus in the disposal condition with the pan structure in the disposal position, according to an illustrative embodiment.
(6) FIG. **5** is a schematic second perspective view of the tray apparatus in the disposal condition with the pan structure in the disposal position, according to an illustrative embodiment.
(7) FIG. **6** is a schematic perspective view of a portion of the tray apparatus showing the latch assembly in the latched condition, according to an illustrative embodiment.
(8) FIG. **7** is a schematic perspective view of a portion of the tray apparatus showing the latch assembly in the unlatched condition, according to an illustrative embodiment.
DETAILED DESCRIPTION
(9) With reference now to the drawings, and in particular to FIGS. **1** through **7** thereof, a new system and apparatus for waste material collection from a cutting machine embodying the principles and concepts of the disclosed subject matter will be described.
(10) Cutting machinery, such as machinery for cutting metal sheet or plate, generates waste or other debris by the material that is cut away (e.g., slugs) as well as the material that melts during the cutting operation and then solidifies upon cooling (e.g., slag). The cutting machinery typically has some means for removing the waste from the machine, such as rudimentary trays, but the trays can be difficult and laborious to clear of waste, particularly the waste that has re-solidified in the tray and may adhere together pieces of the cut away material. Breaking up of amalgamations with a pry bar, and manual picking of the pieces out of the tray is not uncommon to empty the tray.
(11) The applicants have recognized that the original equipment waste collection tray may be improved by including structure that facilitates the lifting of tray with equipment such as a fork lift would be beneficial, and may be even more advantageous if the collection tray is able to provide a waste dumping action when the tray is lifted into an elevated positon above a scrap material dumpster.
(12) In one aspect, the disclosure relates to a system **1** for removing waste or unneeded portions of a material from other portions of the material to be further processed or utilized. One example of

such a system is a material cutting system which cuts material, such as a metal is in the form of a sheet or plate, by means of heat using, for example, laser cutting, plasma cutting, flame cutting or plasma cutting. Such cutting technologies typically generate waste not only from the unused or unneeded portions of sheet or plate material cut away from the usable portions of material, but also the molten portions of the material which eventually cool and harden into irregular waste. Cutting systems are typically positioned in an enclosed building structure and fully or partially rest on a floor surface **2**.

(13) In some aspects, the system **1** may include a material cutting machine **10** which is configured to cut material using one of the disclosed technologies or other suitable technology, and operation of the cutting machine **10** may produce waste pieces of the material, typically in the form of cut away material or material bowled by the process that falls away from the cutting plane. Illustratively, the cutting machine **10** may have a peripheral wall **12** which may extend generally upwardly from the floor surface **2**. Machines **10** which are highly suitable for implementing elements of the disclosure may define at least one collecting space **14** which is located below cutting plane of the machine, and often may be located below the cutting machine. In some of the illustrative implementations, the cutting machine **10** is positioned on the floor surface **2** and one or more collecting spaces **14** are at least partially defined by the floor surface. An opening or aperture **16** may be formed in the peripheral wall **12** of the cutting machine and may be in communication with the collecting space or spaces **14** in order to provide access to the space **14**. The aperture **16** may be at least partially defined by the floor surface **2** so that a portion of the aperture is at floor level.

(14) In some aspects, the system **1** may include a waste collection and disposal tray apparatus **20** for collecting waste material from the material cutting machine **10**, although in some implementations the disclosure is directed to the tray apparatus alone. The tray apparatus **20** may be removably positionable in the collecting space **14** of the cutting machine at a location which permits the apparatus **22** receive waste material generated during the cutting operation, usually by operation of gravity on the unsupported waste portions of the material.

(15) In greater detail, the tray apparatus **20** may have an interior **22** for receiving waste falling or dropping from the raw material placed in the cutting machine such as on the cutting plane. The tray apparatus **20** typically has an open top **24** through which waste falling from cutting operations passes into the interior **22** of the tray apparatus **20**, and also has a bottom **26** on which the waste in the interior **22** rests. The bottom **26** is located opposite of the open top **24**. The tray apparatus **20** may be elongated, and have an inboard end **28** and an outboard end **29**, with the inboard end being inserted through the aperture **16** when the tray apparatus is moved into the collecting space **14** of the machine **10**.

(16) In some of the most significant implementations of the disclosure, the tray apparatus **20** has a collection condition (see, e.g., FIGS. **1** and **2**) and a disposal condition (see, e.g., FIGS. **3** and **4**). The collection condition of the tray apparatus may be characterized by the apparatus **20** being configured to hold waste pieces of material, and the disposal condition may be characterized by the tray apparatus being configured to dump or dispense or otherwise empty waste from the apparatus **20**. Also, another feature, which is highly useful in combination with the conditions of the tray apparatus, is a pair of fork pockets **30**, **32** that are formed by the apparatus **20**. The fork pockets **30**, **32** provide a place of insertion of a portion of a lift fork so that a fork or forks of equipment, such as a forklift, may be utilized to lift and lower the tray apparatus with respect to the floor surface, particularly when waste material is desired to be dumped or emptied from the interior **22** of the apparatus **20**.

(17) In general, the tray apparatus **20** may comprise a base structure **34** and a pan structure **36**. The pan structure **36** is mounted on the base structure **34**, and is movable with respect to the base structure. The pan structure **36** may be movable between a collection position (see, e.g., FIGS. **1** and **2**) and a disposal position (see, e.g., FIGS. **3** and **4**) with respect to the base structure. The

collection position of the pan structure **36** may be characterized by the pan structure having an orientation extending substantially parallel to the base structure **34**. The disposal condition may be characterized by the pan structure having an orientation which is rotated or tilted with respect to the base structure out of the collection position, and may feature the pan structure being angled with respect to the base structure. In some preferred embodiments, the pan structure **36** is pivotally mounted on the base structure **34** at a pivot joint. Illustratively, the pan structure **36** is pivotable on the base structure **34** at a location that is medial of the inboard **28** and outboard **29** ends of the tray apparatus.

(18) In some embodiments, the base structure **34** may include a base wall **40** which is located at the inboard end **29** of the tray apparatus, and may have an entry opening **42** for each of the fork pockets. The base structure **34** may further include a pair of pocket forming elements **44**, **46** which each form one of the fork pockets **30**, **32**. Each of the pocket forming elements **44**, **46** may be tubular to receive a fork of lifting equipment, each element **44**, **46** may be associated with one of the entry openings **42** in the base wall **40**.

(19) In embodiments, the pan structure **36** may include a perimeter wall **50** which extends from the bottom **26** to the open top **24** of the tray apparatus, and about the interior **22**. The perimeter wall **50** may have opposite side portions **52**, **54** and opposite end portions **56**, **58**. Illustratively, the perimeter wall **50** may be substantially rectangular in shape. The pan structure **36** may further include a bottom wall **60** which extends across at least a portion of the bottom of the tray apparatus to generally close the bottom of the interior **22**. The bottom wall **60** may have an underside **62** oriented opposite of the open top **24**.

(20) In some embodiments, the bottom wall **50** may include a pair of inverted channel sections **64**, **66**, with each of the inverted channel sections defining a channel **68** on the underside **62** of the bottom wall for receiving one of the respective pocket forming elements **44**, **46** of the base structure. Each of the inverted channel sections **64**, **66** may extend from the inboard end **28** of the tray apparatus **20** toward the outboard end **29** of the apparatus, and each of the inverted channel sections may extend from an open end **70** at the inboard end to a blind end **72** opposite of the open end, which may be generally medial between the inboard and outboard ends.

(21) The bottom wall **50** may further include a plurality of cavity forming sections **74**, **76** with each of the cavity forming sections defining a cavity **80** on the underside **62** of the bottom wall. Each of the cavity forming sections **74**, **76** may be located toward one of the ends of the tray apparatus, with one of the cavities being an inboard cavity **78** and another one of the cavities being an outboard cavity **80**. In some embodiments, the outer cavity forming section **76** may include an inclined section **82** located toward the outboard end of the pan structure, and the upper surface of the inclined section may be inclined upward from the bottom **26** to the open top **24** at the outboard end of the pan structure to facilitate movement of waste hold of the interior **22** of the apparatus **20**, particularly when the tray apparatus is moved from the collection condition to the disposal condition.

(22) Optionally, the pan structure **36** may include at least one rib **84** which extends upwardly from the bottom wall **60** toward the open top **24**. Illustratively, the rib **84** may extend substantially parallel to the side portions **52**, **54** of the perimeter wall, and may extend from one of the inverted channel sections **64**, **66** of the bottom wall to the inclined section **82** of the bottom wall. In some referred embodiments, a pair of the ribs **84**, **86** are provided and are laterally spaced from each other, and the cited portions **52**, **54**, in the interior **22** of the tray apparatus.

(23) The tray apparatus **20** may further include a securing mechanism **90** which is configured to secure the base structure **34** and the pan structure **36** in the collection position. In some embodiments, the securing mechanism **90** may comprise a latch assembly **92** which is movable between a latched condition (see, e.g., FIG. **6**) and an unlatched condition (see, e.g., FIG. **7**), with the latched condition being characterized by the assembly **92** securing the base **34** and pan **36** structures in the collection position and the unlatched condition being characterized by the

assembly **92** permitting the pan structure **36** to move with respect to the base structure **34** into the disposal position.

(24) In the illustrative embodiments, the latch assembly **92** includes a latch post **94** which is mounted on the pan structure and extends or protrudes from the inboard end of the pan structure. The latch assembly **92** may include a latch member **96** for selectively engaging the latch post **94**. The latch member **96** may be mounted on the base structure, such as on the base wall **40** of the base structure. The latch member is movable with respect to the base structure between a latched position (see, e.g., FIG. **6**) corresponding to the latch condition and an unlatched position (see, e.g., FIG. **7**) corresponding to the unlatched condition of the securing mechanism. The latch member **96** may include a hook portion **98** forming a hook to engage the latch post **94**, and the hook portion may be fastened by a fastener **100** to the base structure in a manner such that the hook portion is pivotable on the fastener. The hook portion **98** may have a throat **102** for receiving a portion of the latch post **94**. The latch member **96** may also include a handle portion **104** which extends from the hook portion **98** to provide a handhold with enhanced leverage for moving the latch member.

(25) The latch assembly **92** may also include a lock member **106** for selectively locking the latch member **96** in the latched position. The lock member **106** may be pivotally mounted on the hook portion **98**, and may have a locked position (see, e.g., FIG. **6**) and an unlocked position (see, e.g., FIG. **7**). The locked position may be characterized by the lock member blocking movement of the latch post **94** out of the throat **102** of the hook portion. The unlocked position may be characterized by the lock member **106** being positioned so as not to lock movement of the latch post out of the throat of the hook portion. Additionally, the latch assembly may include a biasing element **108** which is configured to bias the movement of the latch member **96** toward the latched position, and illustratively may comprise a tension spring.

(26) The tray apparatus **20** may further include a plurality of wheel assemblies **110**, **112**, **114**, **116** which are configured to facilitate movement of the apparatus **20** across the floor surface **2**, particularly when the tray apparatus is to be moved into our out of the collecting space **14** of the machine **10**. The wheel assemblies **110**, **112**, **114**, **116** may be mounted on the pan structure **36**, and may be located on the underside **62** of the bottom wall of the pan structure. Each of the wheel assemblies may be positioned in one of the cavities **78**, **80** on the underside of the bottom wall, and may be arranged such that a wheel assembly is located toward each of the corners of the pan structure.

(27) It should be appreciated that in the foregoing description and appended claims, that the terms “substantially” and “approximately,” when used to modify another term, mean “for the most part” or “being largely but not wholly or completely that which is specified” by the modified term.

(28) It should also be appreciated from the foregoing description that, except when mutually exclusive, the features of the various embodiments described herein may be combined with features of other embodiments as desired while remaining within the intended scope of the disclosure.

(29) In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated.

(30) With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of the disclosed embodiments and implementations, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art in light of the foregoing disclosure, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by the present disclosure.

(31) Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosed subject matter to the exact construction and operation shown

and described, and accordingly, all suitable modifications and equivalents may be resorted to that fall within the scope of the claims.

Claims

1. A system for collecting and disposing waste from a material cutting machine, the system comprising: a waste collection and disposal tray apparatus at least partially insertable into the material cutting machine to receive descending waste, the tray apparatus having an interior for receiving waste from the material cutting machine, the tray apparatus having an open top and a bottom opposite of the open top, the tray apparatus having a collection condition characterized by the tray apparatus being configured to hold waste and a disposal condition characterized by the tray apparatus being configured to dump waste from the tray apparatus, the tray apparatus comprising: a base structure; and a pan structure; wherein the pan structure is pivotally mounted on the base structure at a pivot joint connection to permit the pan structure to pivot with respect to the base structure to move between a collection position corresponding to the collection condition and a disposal position corresponding to the disposal condition; wherein the tray apparatus defines a pair of fork pockets with each fork pocket being configured to receive a portion of a lift fork to facilitate lifting of the tray apparatus by equipment having lift forks.
2. The system of claim 1 wherein the collection position of the tray apparatus is characterized by the pan structure extending substantially parallel to the base structure and the disposal condition of the tray apparatus is characterized by the pan structure being rotated with respect to the base structure out of the collection position.
3. The system of claim 2 wherein the disposal condition of the tray apparatus is characterized by the pan structure being angled with respect to the base structure.
4. The system of claim 1 wherein the base structure includes a pair of pocket forming elements each forming one of the fork pockets, each of the pocket forming elements being tubular.
5. The system of claim 4 wherein the base structure includes a base wall having an entry opening formed therein which is associated with each of the fork pockets.
6. The system of claim 4 wherein the pan structure includes a bottom wall and a perimeter wall extending upwardly from the bottom wall toward the open top, the bottom wall having an underside; and wherein the bottom wall includes a pair of inverted channel sections, each of the inverted channel sections defining a channel on the underside of the bottom wall for receiving one of the pocket forming elements of the base structure.
7. The system of claim 1 wherein the pan structure includes a bottom wall defining a pair of channel sections each receiving at least a portion of a respective one of the fork pockets, each of the fork pockets being joined to a respective one of the channel sections by a said pivot joint connection, each of the pivot joint connections being located in the respective channel section.
8. The system of claim 1 wherein the pivot joint connection of the pan structure and the base structure is located at a fixed location medial of inboard and outboard ends of the tray apparatus.
9. The system of claim 1 wherein the tray apparatus additionally includes a securing mechanism configured to secure the pan structure in the collection position with respect to the base structure.
10. The system of claim 9 wherein the securing mechanism comprises a latch assembly movable between a latched condition characterized by securing the pan structure in the collection position and an unlatched condition characterized by permitting the pan structure to move out of the collection position toward the disposal position.
11. The system of claim 1 wherein the pan structure includes a bottom wall and a perimeter wall extending upwardly from the bottom wall toward the open top, the bottom wall having an underside, the pan structure being elongated between opposite ends; and wherein the bottom wall includes a plurality of cavity forming sections each forming a cavity on the underside of the bottom wall, each of the cavities being positioned toward one of the ends of the pan structure.

12. The system of claim 11 wherein the pan structure includes a plurality of wheel assemblies configured to facilitate movement of the tray apparatus across the floor surface, each of the wheel assemblies being positioned in one of the cavities on the underside of the bottom wall.
13. The system of claim 1 wherein the pan structure includes a bottom wall and a perimeter wall extending upwardly from the bottom wall toward the open top, the pan structure being elongated between opposite inboard and outboard ends; wherein the bottom wall includes an inclined section, an upper surface of the inclined section being inclined upward from the bottom to the open top at the outboard end of the pan structure toward one end of the pan structure to facilitate movement of waste out of the interior of the tray apparatus when the pan structure is moved from the collection position toward the disposal position.
14. A system for collecting and disposing waste from a material cutting machine, the system comprising: a waste collection and disposal tray apparatus at least partially insertable into the material cutting machine to receive descending waste, the tray apparatus having an interior for receiving waste from the material cutting machine, the tray apparatus having an open top and a bottom opposite of the open top, the tray apparatus having a collection condition characterized by the tray apparatus being configured to hold waste and a disposal condition characterized by the tray apparatus being configured to dump waste from the tray apparatus, the tray apparatus comprising: a base structure having a pair of fork pockets with each fork pocket being configured to receive a portion of a lift fork to facilitate lifting of the tray apparatus by equipment having lift forks; and a pan structure, a pivot joint connection pivotally mounting the pan structure on the base structure to permit the pan structure to pivot with respect to the base structure to move between a collection position corresponding to the collection condition and a disposal position corresponding to the disposal condition; wherein the pan structure forms a pair of channels on an underside of the pan structure, each of the channels receiving one of the fork pockets when the pan structure is in the collection position, each of the fork pockets being pivoted substantially out of a respective channel when the pan structure is in the disposal position.
15. The system of claim 14 wherein the pivot joint connection connects each of the fork pockets of the base structure to a respective one of the channels formed by the pan structure.
16. A system for collecting and disposing waste from a material cutting machine, the system comprising: a material cutting machine configured to cut material, operation of the cutting machine producing waste pieces of the material, the cutting machine defining at least one collecting space located below the cutting machine; and a waste collection and disposal tray apparatus at least partially inserted into the at least one collecting space of the material cutting machine to receive descending waste, the tray apparatus being removable from the at least one collecting space, the tray apparatus having an interior for receiving waste from the material cutting machine, the tray apparatus having an open top and a bottom opposite of the open top, the tray apparatus having a collection condition characterized by the tray apparatus being configured to hold waste and a disposal condition characterized by the tray apparatus being configured to dump waste from the tray apparatus, the tray apparatus comprising: a base structure having a pair of fork pockets with each fork pocket being configured to receive a portion of a lift fork to facilitate lifting of the tray apparatus by equipment having lift forks; and a pan structure, a pivot joint connection pivotally mounting the pan structure on the base structure to permit the pan structure to pivot with respect to the base structure to move between a collection position corresponding to the collection condition and a disposal position corresponding to the disposal condition; wherein the pan structure forms a pair of channels on an underside of the pan structure, each of the channels receiving one of the fork pockets when the pan structure is in the collection position, each of the fork pockets being pivoted substantially out of a respective channel when the pan structure is in the disposal position.